$$P = nm\overline{v_x^2} \quad (9.5)$$

Since the gas is assumed to move in random direction, it has no preferred direction of motion (the effect of gravity on the molecules is neglected). It implies that the molecule has same average speed in all the three direction. So, $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$. The mean square speed is written as

$$\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2} = 3\overline{v_x^2}$$

$$\overline{v_x^2} = \frac{1}{3}\overline{v^2}$$

Using this in equation (9.5), we get

$$P = \frac{1}{3}nm\overline{v^2} \text{ or } P = \frac{1}{3}\frac{N}{V}m\overline{v^2} \quad (9.6)$$

as $\left[n = \frac{N}{V}\right]$

The following inference can be made from the above equation. The pressure exerted by the molecules depends on

(i) **Number density** $n = \frac{N}{V}$. It implies that if the number density increases then pressure will increase. For example when we pump air inside the cycle tyre or car tyre essentially the number density increases and as a result the pressure increases.

(ii) **Mass of the molecule** Since the pressure arises due to momentum transfer to the wall, larger mass will have larger momentum for a fixed speed. As a result the pressure will increase.

(iii) **Mean square speed** For a fixed mass if we increase the speed, the average speed will also increase. As a result the pressure will increase.

For simplicity the cubical container is taken into consideration. The above result is true for any shape of the container as the area A does not appear in the final expression (9.6). Hence the pressure exerted by gas molecules on the wall is independent of area of the wall (A).

9.2.2 Kinetic interpretation of temperature

To understand the microscopic origin of temperature in the same way,

Rewrite the equation (9.6)

$$P = \frac{1}{3}\frac{N}{V}m\overline{v^2}$$

$$PV = \frac{1}{3}Nm\overline{v^2} \quad (9.7)$$

Comparing the equation (9.7) with ideal gas equation $PV=NkT$,

$$NkT = \frac{1}{3}Nm\overline{v^2}$$

$$kT = \frac{1}{3}m\overline{v^2} \quad (9.8)$$

Multiply the above equation by 3/2 on both sides,

$$\frac{3}{2}kT = \frac{1}{2}m\overline{v^2} \quad (9.9)$$

R.H.S of the equation (9.9) is called average kinetic energy of a single molecule ($\overline{KE}$).

The average kinetic energy per molecule

$$\overline{KE} = \epsilon = \frac{3}{2}kT \quad (9.10)$$

Equation (9.9) implies that the temperature of a gas is a measure of the average translational kinetic energy per molecule of the gas.

Note Compare this with the definition of temperature studied in lower classes: Temperature is the degree of hotness or coldness!

Equation 9.10 is a very important result from kinetic theory of gas. We can infer the following from this equation.

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